

Lab 12 –

Introduction to Random Module

Programming Exercise:

Q.1 Create a list of different Cities of Pakistan and by using random module Shuffle that list.

Source Code:

```
1  import random
2
3  # List of different cities in Pakistan
4  cities = ["Karachi", "Lahore", "Islamabad", "Rawalpindi", "Peshawar",
5           "Quetta", "Multan", "Faisalabad", "Sialkot", "Gujranwala"]
6
7  # Shuffle the list using random.shuffle()
8  random.shuffle(cities)
9
10 # Print the shuffled list
11 print(cities)
12
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\main 3.py"
['Gujranwala', 'Rawalpindi', 'Karachi', 'Sialkot', 'Multan', 'Peshawar', 'Faisalabad', 'Lahore', 'Islamabad', 'Quetta']

Process finished with exit code 0
```

Q.2 Define a list, consisting of names of different students, who are applying for a scholarship, pop an item from that list and transfer all items in a new list and apply sample method of random module to give scholarship to two students of new list.

Source Code:

```
1  import random
2
3  # List of students applying for a scholarship
4  students = ["Ali", "Bisma", "Haris", "Dawood", "Eshaal", "Farheen", "Ghazi"]
5
6  popped_student = students.pop(0) # Removes 'Ali' from the list
7  print("Popped student:", popped_student)
8
9  # Transfer the remaining items to a new list
10 new_list = students.copy()
11
12 # Use random.sample() to select 2 students for the scholarship
13 scholarship_students = random.sample(new_list, k= 2)
14
15 # Print the students awarded the scholarship
16 print("Students awarded the scholarship:", scholarship_students)
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp
```

```
Popped student: Ali
```

```
Students awarded the scholarship: ['Ghazi', 'Farheen']
```

```
Process finished with exit code 0
```

Q.3 Define two lists dice1 and dice2, holding numbers from 1 to 6. Design a python program that start playing with two dices for two players

Source Code:

```
1  import random
2
3  # Lists representing the numbers on two dice
4  dice1 = [1, 2, 3, 4, 5, 6]
5  dice2 = [1, 2, 3, 4, 5, 6]
6
7
8  # Function to simulate a dice roll for a player
9  def roll_dice(dice): 2 usages
10     return random.choice(dice)
11
12
13 # Function to play a round between two players
14 def play_round(player1, player2): 1 usage
15     roll1 = roll_dice(dice1)
16     roll2 = roll_dice(dice2)
17
18     print(f"{player1} rolled a {roll1}")
19     print(f"{player2} rolled a {roll2}")
20
21     if roll1 > roll2:
22         print(f"{player1} wins this round!")
23         return player1
24     elif roll2 > roll1:
25         print(f"{player2} wins this round!")
26         return player2
27     else:
28         print("It's a tie!")
29         return "Tie"
30
31
32 # Main function to play the game
33 def play_game(): 1 usage
34     player1 = "Player 1"
35     player2 = "Player 2"
36
37     rounds = int(input("Enter number of rounds to play: ")) # Number of rounds in the game
38     player1_score = 0
39     player2_score = 0
```

```

40
41     for round_num in range(1, rounds + 1):
42         print(f"\n--- Round {round_num} ---")
43         winner = play_round(player1, player2)
44
45         if winner == player1:
46             player1_score += 1
47         elif winner == player2:
48             player2_score += 1
49
50     # Final score after all rounds
51     print(f"\nFinal Scores: {player1}: {player1_score}, {player2}: {player2_score}")
52
53     if player1_score > player2_score:
54         print(f"{player1} wins the game!")
55     elif player2_score > player1_score:
56         print(f"{player2} wins the game!")
57     else:
58         print("The game is a tie!")
59
60
61 # Run the game
62 play_game()

```

Output:

C:\Users\hp\PycharmProjects\main\venv\Script

Enter number of rounds to play: 3

```

--- Round 1 ---
Player 1 rolled a 2
Player 2 rolled a 6
Player 2 wins this round!

```

```

--- Round 2 ---
Player 1 rolled a 4
Player 2 rolled a 5
Player 2 wins this round!

```

```

--- Round 3 ---
Player 1 rolled a 2
Player 2 rolled a 2
It's a tie!

```

```

Final Scores: Player 1: 0, Player 2: 2
Player 2 wins the game!

```

Process finished with exit code 0

